

HEALTHCARE SA – DATA REQUIREMENTS AND ANALYSIS SPECIFICATION

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DISCLAIMER: SYNTHETIC DATA

This dataset is 100% synthetic and it is created for educational/portfolio purposes only.

It does not contain real patient data, it is not from any SA hospital or Department of Health and any resemblance to real persons is coincidental.

Project Overview

Analysis of patient admissions data for South African healthcare facilities to support operational reporting, resource planning and clinical insights.

Dataset: 3 Files

- Raw data
- Clean data
- Post clean data

Business Objectives

- Provide clean, trusted patient dataset for BI reporting
- Track admissions by Province, Diagnosis and Date
- Enable demographic analysis (Age, Gender & Blood type)
- Ensure POPIA compliance

Data Dictionary

- Patient-ID: Unique number for each patient
- Full name: Patient name and surname
- Age: How old is the patient
- Gender: Male or Female
- Province: Where they were admitted
- Blood type: the blood type of each patient
- Diagnosis: why they came
- Admission date: when they came

Problems to identify and fix

- Missing values
- Duplicate patient records

- Invalid ages
- Inconsistent gender values
- Inconsistent diagnosis names
- Invalid blood type formats
- Incorrect data types
- Outliers

Data Quality Assessment

- Column naming
- Date format
- Duplicate records

Analysis Requirements

- Total admissions by province
- Admissions trend by month
- Diagnosis distribution
- Age group
- Gender vs diagnosis
- Blood type distribution

Dashboards

- Power BI charts

Deliverables

- Cleaned dataset
- SQL queries
- Dashboard picture
- Final analysis report